オウ　イシン
王　一臻

WANG Yizhen

Master | Mechanical Engineering | Hasegawa Lab, The University of Tokyo

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Education

The University of Tokyo — **Oct 2023 – Sep 2025**
Master, Mechanical Engineering, Hasegawa Lab — Tokyo, Japan

- Automatic Optimization of Pulsating Waveform for Drag Reduction in Experiment of Turbulent Pipe Flow
- Built an unattended Bayesian-optimization loop over the rig, reaching **75% maximum drag reduction** and a **40% maximum power saving rate**

  Thesis received the JSAE Graduate Research Encouragement Award

University of Melbourne — **Jan 2019 – Dec 2019**
Master, Mechanical Engineering (Withdrawn) — Melbourne, Australia

- Investigation of energetic motions in rough-wall turbulent boundary layers

  Withdrew on receiving a fully funded PhD offer from the same university; enrolment was deferred under COVID-19 travel and visa restrictions and the offer lapsed in late 2022 under university policy

Shanghai Jiao Tong University — **Sep 2014 – Sep 2018**
Bachelor, Energy and Power Engineering; Minor in Musicology — Shanghai, China

- Dilution effects of DME and ethanol on methane flames
- Dynamic characteristics of hydrogen-cooled generator stator foundations

Work Experience

SOLIZE Partners — **Apr 2026 – Present**
CAE Analysis Engineer — Kanagawa, Japan

- Fluid-structure coupled thermal, combustion-coupled and supercritical-fluid analysis for aerospace propulsion, run in parallel on an HPC cluster
- Delivered optimization PoCs: piping support placement (heat loss, thermal stress, seismic natural frequency) and coupled fluid-thermal-structural with topology optimization
- Cross-validation across ANSYS Fluent / OpenFOAM / ToffeeX; built the internal OpenFOAM computing environment (Slurm)

Independent Academic Tutor — **2020 – 2023**
AP / IB / A-Level Mathematics and Physics — Shanghai, China

- Taught students preparing for overseas university admission while reaching business-level Japanese and securing University of Tokyo admission

Skills

- **Mechanical Engineering**
  Tools: SolidWorks, OpenFOAM, ANSYS
  Coursework: Advanced Heat Transfer, Numerical Thermofluid Engineering, Advanced Fluid Engineering, Numerical Methods
- **Electrical Engineering**
  Coursework: Circuit Theory, Visualization of PLC Control, Measurement Principles and Technologies, Instrumentation Concepts and System/Control Modeling
- **Machine Learning & Optimization**
  Methods/Coursework: Bayesian Optimization, Brain-Inspired Information Processing Mechanics, Neuro Intelligence
- **Miscellaneous**
  Programming: Python, MATLAB, C/C++, Emacs Lisp
  Tools: Microsoft Office, LaTeX
  Systems: GNU/Linux, openWRT, BSD

Languages

- Mandarin Chinese: Native
- English: Native Level
- Japanese: JLPT N1 (Business Level)

Interests

- Classical Music: Amateur Violist
- PC DIY: Server Administrator
- Detective Novels and Science Fiction